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Gradient damage models applied to dynamic fragmentation of brittle materials

Arthur Geromel Fischer · Jean-Jacques Marigo

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Abstract This paper is devoted to the use of gradient damage models in a dynamical context. After the setting of the general dynamical problem using a variational approach, one focuses on its application to the fragmentation of a brittle ring under expansion. Although the 1D problem admits a solution where the damage field remains uniform in space, numerical simulations show that the damage field localizes in space at a certain time and then a fragmentation of the ring rapidly occurs. To understand this phenomenon from a theoretical point of view, one develops a stability analysis of the homogeneous response by studying the growth of small perturbations. A dimensional analysis shows that the problem essentially depends on two dimensionless parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$, $\tilde{\ell}$ being related to the characteristic length present in the damage model and $\tilde{\dot{\varepsilon}}_0$ to the applied expansion rate. Then, since the product $\tilde{\ell}\tilde{\dot{\varepsilon}}_0$ is small in practice, the problem of stability is solved in a closed form by using asymptotic expansions. The comparison with the numerical results allows us to conclude that the time at which the damage localizes and the number of fragments are really governed by the growth of the imperfections. To conclude, a numerical simulation of the fragmentation of a 2D ring is presented.

A. Geromel Fischer · J.-J. Marigo (✉)
Laboratoire de Mécanique des Solides, École Polytechnique, 91128 Palaiseau, France
e-mail: jean-jacques.marigo@polytechnique.edu

A. G. Fischer
e-mail: arthur.geromel-fischer@polytechnique.edu

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1 Introduction

Gradient damage models (Pham and Marigo 2010a, b; Pham et al. 2011b), which are also called phase field models in the literature (Hakim and Karma 2009; Miehe et al. 2015), have known a great success in the last decade by virtue of their ability to treat all the process of fracture of bodies, from the nucleation of cracks to the final failure. Their first interest, from a theoretical point of view, is that they contain a critical stress which governs the nucleation of cracks (Pham and Marigo 2010b, 2011; Pham et al. 2011a), contrarily to Griffith's model (Griffith 1921; Francfort and Marigo 1998) which allows infinite stresses and hence is not able to account for nucleation. Another benefit is that they contain a regularized term which prevents damage localization in bands of zero thickness and, hence, require that some energy is spent to create a crack, in contrast to local (non reguralized) damage models (Pham et al. 2011a; Pham and Marigo 2011). From a numerical point of view, their main convenience is that they allow the full problem, from nucleation to failure, to be treated using classical finite element methods and without a priori assumptions on the crack path (Bourdin et al. 2000). Everything is now well-understood and their use is well controlled in a quasi-

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static setting for brittle materials, see (de Borst et al. 1996; Francfort and Marigo 1998; Bourdin et al. 2000, 2008; Lorentz and Benallal 2005; Hakim and Karma 2009; Comi 2001; Miehe et al. 2015). It is now possible to account for very complex crack patterns and paths, like in the case of a thermal shock problem (Bourdin et al. 2014; Sicsic et al. 2014). Their link with Griffith's theory is now well established, either by proving Gamma-convergence results (Braides 2002; Dal-Maso and Toader 2001; Bourdin et al. 2008) or by using mechanical arguments (Pham et al. 2011a; Pham and Marigo 2011). Their extension to ductile fracture by introducing plastic strains and their coupling with damage is in progress and the first attempts are very promising (Alessi et al. 2014, 2015; Ambati et al. 2015; Miehe et al. 2015).

But the most important challenge is now to extend the use of gradient damage models in a dynamical setting (both for brittle and ductile fracture) and to see whether their great qualities observed in quasi-static remain true in dynamics. Some first attempts have been made (Bourdin et al. 2011; Borden et al. 2012; Li and Marigo 2017; Li et al. 2016), but several fundamental issues remain unsolved. From a theoretical point of view, the first issue is how to formulate the damage evolution problem in dynamics. Indeed, the variational approach as it was developed in quasi-static (Bourdin et al. 2008) must be changed in dynamics because one can no longer use the energetic stability criterion (Benallal and Marigo 2007; Pham and Marigo 2010a, b; Pham et al. 2011b). Some new principles have been proposed (Larsen 2010), but the most natural idea seems to adapt the classical principle of least action in the context of damage like in Li and Marigo (2017) and Li et al. (2016). We will take this approach in the present paper. Once the formulation has been chosen, the next issue is to establish some general qualitative properties of these gradient damage models in dynamics, in particular the questions of uniqueness of the solution and its stability with respect to small perturbations. Another question is to see whether these models are able to predict in a correct way the phenomenon of multiple fragmentation which is observed in the case of high strain rate both for brittle and ductile materials (Ravi-Chandar 1998; Zinszner et al. 2015). Very little has been made at the present time on those fundamentals questions. We propose here to address them in the fundamental problem of the fragmentation of a brittle ring submitted to an extension at high strain rate. In the past, many works have been devoted to that problem either by using simplified model (Grady 1982; Glenn and Chudnovsky 1986) or by solving a full dynamical problem with the help of numerical simulation based on cohesive zone models (Miller et al. 1999; Pandolfi et al. 1999; Drugan 2001; Molinari et al. 2007; Levy and Molinari 2010), but to our knowledge it is the first time where that problem is addressed with phase field models.

Specifically the paper is organized as follows. One first briefly recall in Sect. 2 the main ingredients of a gradient damage model, before formulating the general evolution problem in a dynamical setting. Section 3 is devoted to the setting of the 1D version of the problem of the expansion of a brittle ring. One shows in particular that the problem admits a solution where the damage field remains uniform in space (and grows in time). The study of the fragmentation of the 1D ring is made in Sect. 4. Numerical simulations are firstly presented, showing that the damage field localizes in space at a certain time which leads to a rapid fragmentation of the ring. To understand this phenomenon from a theoretical point of view, one develops a stability analysis of the homogeneous response by studying the growth of small perturbations, as in Mercier and Molinari (2003), Rodríguez-Martinez et al. (2013), Vaz-Romero et al. (2017) and Ravi-Chandar and Triantafyllidis (2015). A dimensional analysis shows that the problem essentially depends on two dimensionless parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$, $\tilde{\ell}$ being related to the characteristic length present in the damage model and $\tilde{\dot{\varepsilon}}_0$ to the applied expansion rate. Then, using the fact that in practice the product $\tilde{\ell}\tilde{\dot{\varepsilon}}_0$ is small, one is able to solve the problem of stability in a closed form by using asymptotic expansions. The comparison with the numerical results allows us to conclude that the time at which the damage localizes and the number of fragments are really governed by the growth of the imperfections. Then a comparison is made with the simplified approaches developed by Grady (1982) or Glenn and Chudnovsky (1986). Finally in Sect. 5 numerical simulation of complete 3D ring are presented and compared to those obtained in the 1D setting.

The used notation are briefly summarized here. In a 3D setting the stress is denoted by $\boldsymbol{\sigma}$, the strain by $\boldsymbol{\varepsilon}$, the displacement by $\mathbf{u}$ and the rigidity tensor by $\mathbf{E}$. The scalar product of two vectors or two tensors $\mathbf{a}$ and $\mathbf{b}$ will be denoted by a dot, $\mathbf{a} \cdot \mathbf{b}$. The strain is the symmetrical part of the gradient of the displacement, what reads as

$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u}+\nabla^T\mathbf{u})$. When working in a 1D setting, we simply denote the Young's Modulus by E, the stress by σ and the strain by ε. The spatial derivative is denoted with a prime whereas the time derivative is indicated by a dot: $u' := \partial u/\partial x$, $\dot{u} := \partial u/\partial t$.

2 Gradient damage models

In this section, we briefly explain the construction of the so-called "gradient damage models" and its extension to dynamic loadings. We consider that the material is brittle elastic, and that there are no other dissipative phenomena. We consider only the case of small strains theory and isotropic material. For a more detailed construction of these models, see (Bourdin et al. 2008; Pham and Marigo 2010a, b; Pham et al. 2011b).

2.1 Quasi-static formulation

Let us first recall the main ingredients of the modeling of damage in a quasi-static setting.

1. *The damage parameter.* We assume that the damage can be represented by a scalar $\alpha\in[0,1]$. When $\alpha = 0$ the material is sound and when $\alpha = 1$ the material is completely broken.
2. *Progressive loss of rigidity.* The rigidity tensor $\mathbf{E}(\alpha)$ is a function of α, the material becomes less rigid when α increases and $\mathbf{E}(1) = 0$ (no more stiffness when the material is broken). It is important to notice that, for a fixed damage value, the stress–strain relation is supposed to be linear:

$$\boldsymbol{\sigma} = \mathbf{E}(\alpha)\boldsymbol{\varepsilon} \tag{1}$$

   which means in particular that one does not distinguish tension and compression. However, in the case of a ring expansion, this limitation is not really important.
3. *Irreversibility condition.* Damage is irreversible, that is, it can only grow in time:

$$\dot{\alpha} \geq 0. \tag{2}$$

4. *Damage evolution law in the case of a non regularized model.* To specify the conditions for damage evolution, we first consider the case of a so-called *local* damage model. Its regularization by damage gradient terms will be introduced in the next paragraph. The evolution law is based on the notion of critical elastic energy release rate, like Griffith's criterion (Griffith 1921) in fracture mechanics. That criterion, which can be deduced from Drucker–Ilyushin postulate (Marigo 1989), will allow us to obtain a variational formulation of the damage evolution problem. Specifically, since the elastic energy is given by

$$\psi(\boldsymbol{\varepsilon}, \alpha) = \frac{1}{2}\mathbf{E}(\alpha)\boldsymbol{\varepsilon}\cdot\boldsymbol{\varepsilon}, \tag{3}$$

the damage evolution is governed by the following conditions

$$\begin{aligned} &-\frac{1}{2}\mathbf{E}'(\alpha)\boldsymbol{\varepsilon}\cdot\boldsymbol{\varepsilon} \leq k(\alpha), \\ &\begin{cases} \dot{\alpha} = 0 & \text{if } -\frac{1}{2}\mathbf{E}'(\alpha)\boldsymbol{\varepsilon}\cdot\boldsymbol{\varepsilon} < k(\alpha) \\ \dot{\alpha} \geq 0 & \text{if } -\frac{1}{2}\mathbf{E}'(\alpha)\boldsymbol{\varepsilon}\cdot\boldsymbol{\varepsilon} = k(\alpha) \end{cases} \end{aligned} \tag{4}$$

where $k(\alpha)$ is a positive function of α representing the critical elastic energy release rate.
Consequently, introducing the primitive of $k(\alpha)$ which vanishes at $\alpha = 0$,

$$w(\alpha) = \int_0^\alpha k(\beta)d\beta,$$

it turns out that, by virtue of (4), the strain work done $W := \int_0^1 \boldsymbol{\sigma}_t\cdot\dot{\boldsymbol{\varepsilon}}_t dt$ during a process where the state $(\boldsymbol{\varepsilon}_t, \alpha_t)$ of the volume element evolves from $(\mathbf{0}, 0)$ to $(\boldsymbol{\varepsilon}, \alpha)$ is a function of the final state only, given by

$$W(\varepsilon, \alpha) = \psi(\varepsilon, \alpha) + w(\alpha). \tag{5}$$

Therefore, $w(\alpha)$ can be interpreted as the energy dissipated when the damage parameter grows from 0 to α. Moreover, the damage law can be written in terms of the energy density function W only. Specifically, the stress–strain relation reads as

$$\boldsymbol{\sigma} = \frac{\partial W}{\partial \boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon}, \alpha) \tag{6}$$

and the damage evolution becomes

$$\begin{cases} \textit{Irreversibility condition}: & \dot{\alpha} \geq 0, \\ \textit{Damage criterion}: & \dfrac{\partial W}{\partial \alpha}(\varepsilon, \alpha) \geq 0, \\ \textit{Consistency relation}: & \dfrac{\partial W}{\partial \alpha}(\varepsilon, \alpha)\dot{\alpha} = 0. \end{cases} \tag{7}$$

Let us recall that the damage criterion can be also interpreted as a first order stability condition, and the consistency relation as an energy balance, see (Pham and Marigo 2010a; Alessi et al. 2015) for details.

5. *Introduction of the damage gradient.* It is now generally admitted that local softening damage models are not viable (Alessi et al. 2015; Pham et al. 2011a, b; Pham and Marigo 2011) as they allow damage localization in infinitely thin bands. To solve this problem, a regularizing term is used. The main idea is to add a *characteristic length* in order to penalize sharp damage profiles and solve the problem of localization in thin bands and fracture without energy dissipation.
Accordingly, we introduce a damage gradient term into the energy density W which reads now as

$$W(\boldsymbol{\varepsilon}, \alpha, \nabla\alpha) = \psi(\boldsymbol{\varepsilon}, \alpha) + w(\alpha) + \frac{1}{2} w_1 \ell^2 \nabla\alpha \cdot \nabla\alpha, \tag{8}$$

where ℓ is a characteristic length and $w_1 := w(1) > 0$ can be seen as a normalization constant. Because of the presence of this "non local" term, the damage evolution law must be formulated first at the level of the whole structure.
Thus, let us consider a body whose reference configuration is the open set Ω of $\mathbb{R}^n$. Starting from an initial damage state α_0 at time $t = 0$, the body is submitted to a loading process characterized, at each time t, by the body forces $\mathbf{f}_t$, the surfaces forces $\mathbf{F}_t$ on the part $\partial_F \Omega$ of its boundary and a prescribed displacement $\mathbf{U}_t$ on the complementary part $\partial_U \Omega$ of its boundary. Accordingly, at time t, the set of kinematically admissible displacement fields is

$$\mathcal{C}_t = \{\mathbf{v} : \mathbf{v} = \mathbf{U}_t \quad \text{on} \quad \partial_U \Omega\},$$

whereas the set of accessible damage state from the current damage set α_t is, by virtue of the irreversibility condition,

$$\mathcal{D}_t = \{\beta : \alpha_t \leq \beta \leq 1 \quad \text{in} \quad \Omega\}.$$

At any pair $(\mathbf{v}, \beta) \in \mathcal{C}_t \times \mathcal{D}_0$ considered as a possible state of the body at time t, one associates the total energy which is defined (at this time) by

$$\mathcal{E}_t(\mathbf{v}, \beta) = \int_\Omega W(\boldsymbol{\varepsilon}(\mathbf{v}), \beta, \nabla\beta) dx - \int_\Omega \mathbf{f}_t \cdot \mathbf{v} dx - \int_{\partial_F \Omega} \mathbf{F}_t \cdot \mathbf{v} dS. \tag{9}$$

(Of course, all these fields must be smooth enough so that the energy be finite, but those regularity conditions will not be discussed here.) Under these definitions, the evolution problem can now be formulated in terms of the following three items of irreversibility, stability and energy balance. Specifically, the problem consists on finding $t \mapsto (\mathbf{u}_t, \alpha_t) \in \mathcal{C}_t \times \mathcal{D}_0$ such that, at any time $t > 0$:

$$\begin{cases} \textit{Irreversibility:} & \dot{\alpha}_t \geq 0 \\ \textit{First order stability:} & \mathcal{E}_t'(\mathbf{u}_t, \alpha_t)(\mathbf{v} - \mathbf{u}_t, \beta - \alpha_t) \geq 0, \\ & \quad \forall (\mathbf{v}, \beta) \in \mathcal{C}_t \times \mathcal{D}_t \\ \textit{Energy balance:} & \dfrac{d}{dt}\mathcal{E}_t(\mathbf{u}_t, \alpha_t) = -\displaystyle\int_\Omega \dot{\mathbf{f}}_t \cdot \mathbf{u}_t dx \\ & \quad -\displaystyle\int_{\partial_F \Omega} \dot{\mathbf{F}}_t \cdot \mathbf{u}_t dS + \int_{\partial_U \Omega} \boldsymbol{\sigma}_t \mathbf{n} \cdot \dot{\mathbf{U}}_t dS. \end{cases} \tag{10}$$

In the first order stability condition above, $\mathcal{E}_t'(\mathbf{u}_t, \alpha_t)(\mathbf{v}, \beta)$ denotes the directional derivative of $\mathcal{E}_t$ at $(\mathbf{u}_t, \alpha_t)$ in the direction $(\mathbf{v}, \beta)$. That condition can be reinforced by a local stability condition in order to select the good solutions in the case of multiple solutions. We do not use this extended stability condition here (because it does not work in dynamics and must be replaced by another concept of stability) and the interested reader can refer to Pham et al. (2011b) for its definition and its use. In the energy balance, $\boldsymbol{\sigma}_t$ denotes the current stress field, whereas $\dot{\mathbf{f}}_t$, $\dot{\mathbf{F}}_t$ and $\dot{\mathbf{U}}_t$ denote the rate of the data.
Using standard arguments of calculus of variations and provided that the evolution is sufficiently smooth (both in space and time), one can show that the three items of (10) are satisfied only if:

(a) the stress field $\boldsymbol{\sigma}_t = \mathbf{E}(\alpha_t)\boldsymbol{\varepsilon}(\mathbf{u}_t)$ satisfies the equilibrium equations

$$\operatorname{div}\boldsymbol{\sigma}_t + \mathbf{f}_t = \mathbf{0} \quad \text{in} \quad \Omega, \tag{11}$$

(b) the damage field α_t satisfies the nonlocal damage criterion

$$\frac{1}{2}\mathbf{E}'(\alpha_t)\boldsymbol{\varepsilon}(\mathbf{u}_t) \cdot \boldsymbol{\varepsilon}(\mathbf{u}_t) + w'(\alpha_t) - w_1 \ell^2 \Delta\alpha_t \geq 0 \quad \text{in} \quad \Omega \tag{12}$$

and the nonlocal consistency condition

$$\Big(\frac{1}{2}\mathbf{E}'(\alpha_t)\boldsymbol{\varepsilon}(\mathbf{u}_t)\cdot\boldsymbol{\varepsilon}(\mathbf{u}_t)+w'(\alpha_t)\\ -w_1\ell^2\Delta\alpha_t\Big)\dot{\alpha}_t=0 \quad \text{in} \quad \Omega. \tag{13}$$

Moreover one also obtains natural boundary conditions both for the stress vector $\boldsymbol{\sigma}_t\mathbf{n}$ on $\partial_F\Omega$ and for the normal damage gradient $\partial\alpha_t/\partial n$ on the part of the boundary where the damage is not prescribed, see Pham and Marigo (2010b).

2.2 Dynamic formulation

To obtain the dynamic formulation of the damage evolution problem, one must introduce inertial effects via the kinetic energy. Unfortunately, one can no longer use the first order stability condition, but replace it by a principle of least action like in Li and Marigo (2017). Specifically, let us denote by $\mathcal{K}(\boldsymbol{v})$ the kinetic energy of the body associated with the velocity field $\boldsymbol{v}$,

$$\mathcal{K}(\boldsymbol{v}):=\int_\Omega \frac{\rho}{2}\boldsymbol{v}\cdot\boldsymbol{v}\,dx, \tag{14}$$

where ρ is the mass density. The lagrangian of the body at time t associated with the (admissible) displacement field $\mathbf{v}$, the (admissible) damage field β and the (admissible) velocity field $\boldsymbol{v}$ is given by

$$\mathcal{L}_t(\mathbf{v},\beta,\boldsymbol{v}):=\mathcal{E}_t(\mathbf{v},\beta)-\mathcal{K}(\boldsymbol{v}),$$

where $\mathcal{E}_t(\mathbf{v},\beta)$ is still given by (9). The action associated with the history of the displacement and damage fields $t\mapsto(\mathbf{v}_t,\beta_t)$ during the time interval $[0,T]$ is defined as

$$\mathcal{A}(\mathbf{v},\beta)=\int_0^T \mathcal{L}_t(\mathbf{v}_t,\beta_t,\dot{\mathbf{v}}_t)dt.$$

Finally, the first order stability condition of (10) is now replaced by the following stationary condition for the action ($\mathcal{A}'$ denoting the directional derivative of $\mathcal{A}$):

$$\mathcal{A}'(\mathbf{u},\alpha)(\mathbf{v},\beta)\geq 0 \tag{15}$$

which must hold

(i) for every $t\mapsto\mathbf{v}_t$ such that $\mathbf{v}_0=\mathbf{v}_T=\mathbf{0}$ in Ω, and $\mathbf{v}_t=\mathbf{0}$ on $\partial_U\Omega$ for every $t\in[0,T]$;
(ii) for every $t\mapsto\beta_t$ such that $\beta_t\geq 0$ in Ω for every $t\in[0,T]$.

Now the current total mechanical energy of the body at time t is given by $\mathcal{E}_t(\mathbf{u}_t,\alpha_t)+\mathcal{K}(\dot{\mathbf{u}}_t)$ and includes the current kinetic energy. Accordingly, the energy balance at time t is simply

$$\frac{d}{dt}\Big(\mathcal{E}_t(\mathbf{u}_t,\alpha_t)+\mathcal{K}(\dot{\mathbf{u}}_t)\Big)\\ =-\int_\Omega \dot{\mathbf{f}}_t\cdot\mathbf{u}_t dx-\int_{\partial_F\Omega}\dot{\mathbf{F}}_t\cdot\mathbf{u}_t dS\\ +\int_{\partial_U\Omega}\boldsymbol{\sigma}_t\mathbf{n}\cdot\dot{\mathbf{U}}_t dS \tag{16}$$

and changes from the quasi-static statement only by the presence of the kinetic energy.

Consequently, using standard arguments of calculus of variations and provided that the evolution is sufficiently smooth, one deduces from (2), (15) and (16) that the equilibrium equation (11) is simply replaced by the dynamical equation

$$\mathrm{div}\boldsymbol{\sigma}_t+\mathbf{f}_t-\rho\ddot{\mathbf{u}}_t=\mathbf{0} \quad \text{in} \quad \Omega \tag{17}$$

whereas the damage criterion (12) and the consistency relation (13) remain unchanged.

3 Application to the expansion of 1D ring

3.1 Setting of the problem

We address now the issue of fragmentation of a brittle ring under expansion. We first consider the simplified 1D problem, the 3D version will be studied numerically in the last section of the paper. All the calculations and the numerical simulations will be made with the following damage model (see Pham et al. 2011a; Pham and Marigo 2011 for some comments on its properties):

$$E(\alpha)=(1-\alpha)^2E_0, \qquad w(\alpha)=\frac{\sigma_c^2}{E_0}\alpha \tag{18}$$

where E_0 represents the Young modulus of the sound material and σ_c is interpreted as the maximal stress that the material can sustain. Therefore, the critical strain ε_c at which damage occurs under a monotonic traction test is given by

$$\varepsilon_c=\frac{\sigma_c}{E_0}. \tag{19}$$

The ring of length L and whose reference configuration is the interval $[0,L]$ is submitted to a uniform expansion so that the strain field ε_t at a time t can be written as

$$\varepsilon_t(x) = \dot{\varepsilon}_0 t + \bar{\varepsilon}_t(x) \quad \text{with} \quad \int_0^L \bar{\varepsilon}_t(x)dx = 0,$$

where $\dot{\varepsilon}_0$ denotes the given strain rate which will play the role of the loading parameter. Therefore, the displacement field u_t at time t can be decomposed as

$$u_t(x) = \dot{\varepsilon}_0 x t + \bar{u}_t(x) \quad \text{with} \quad \bar{u}_t(0) = \bar{u}_t(L) \tag{20}$$

and hence the displacement $\bar{u}_t$ is periodic. The stress field σ_t, the damage field α_t and its gradient α_t' must also be periodic and hence the following relations must hold at any time:

$$\sigma_t(0) = \sigma_t(L), \quad \alpha_t(0) = \alpha_t(L), \quad \alpha_t'(0) = \alpha_t'(L). \tag{21}$$

So, the problem consists in finding $t \mapsto \bar{u}_t$ and $t \mapsto \alpha_t$ which must satisfy at any time $t > 0$ in the domain $(0, L)$ the following conditions

$$\begin{cases} \textit{motion equation}: & \sigma_t' - \rho \ddot{\bar{u}}_t = 0 \\ \textit{stress–strain relation}: & \sigma_t = (1-\alpha_t)^2 E_0(\bar{u}_t' + \dot{\varepsilon}_0 t) \\ \textit{irreversibility condition}: & \dot{\alpha}_t \geq 0 \\ \textit{damage criterion}: & (1-\alpha_t)(\bar{u}_t' + \dot{\varepsilon}_0 t)^2 \\ & \quad -\varepsilon_c^2 + \varepsilon_c^2 \ell^2 \alpha_t'' \leq 0 \\ \textit{consistency relation}: & \Big((1-\alpha_t)(\bar{u}_t' + \dot{\varepsilon}_0 t)^2 \\ & \quad -\varepsilon_c^2 + \varepsilon_c^2 \ell^2 \alpha_t''\Big)\dot{\alpha}_t = 0 \end{cases} \tag{22}$$

at which one must add the periodic conditions (20)–(21) and the initial conditions giving the damage state, the displacement and the velocity fields at time 0.

3.2 The homogeneous response when the ring starts from its sound natural state

Let us assume that, at time 0, the ring is sound and that it starts from its natural configuration with a velocity corresponding with the uniform strain rate $\dot{\varepsilon}_0$. In other words, one assumes that at time 0 the following conditions hold true:

$$\alpha_0(x) = 0, \quad \bar{u}_0(x) = 0, \quad \dot{\bar{u}}_0(x) = 0, \quad \forall x \in [0, L]. \tag{23}$$

Then, one easily checks that the problem (21)–(23) admits a solution where the strain and the damage fields are uniform in space. Specifically, $\bar{u}_t = 0$ for every $t \geq 0$ and the damage field $x \mapsto \alpha_t(x)$ equal everywhere in $[0, L]$ to the following time dependent constant

$$\alpha_t^* = \begin{cases} 0 & \text{when} \quad 0 \leq t \leq \dfrac{\varepsilon_c}{\dot{\varepsilon}_0} \\ 1 - \dfrac{\varepsilon_c^2}{\dot{\varepsilon}_0^2 t^2} & \text{when} \quad t \geq \dfrac{\varepsilon_c}{\dot{\varepsilon}_0} \end{cases} \tag{24}$$

satisfy all the required relations (21)–(23). The associated stress field is also uniform in space and its evolution in time is given by

$$\sigma_t^* = \begin{cases} E_0 \dot{\varepsilon}_0 t & \text{when} \quad 0 \leq t \leq \dfrac{\varepsilon_c}{\dot{\varepsilon}_0} \\ \dfrac{\varepsilon_c^3}{\dot{\varepsilon}_0^3 t^3}\sigma_c & \text{when} \quad t \geq \dfrac{\varepsilon_c}{\dot{\varepsilon}_0} \end{cases} \tag{25}$$

Note that the damage variable tends asymptotically to 1 and the stress to 0 (uniformly in space) when the time goes to infinity.

3.3 The issue of the uniqueness and the stability of the homogeneous response

Let us first remark that the homogeneous response is also solution of the quasi-static evolution, that is the problem where the equation of motion in (22) is replaced by the equilibrium equation $\sigma_t' = 0$ (and the initial kinematical conditions are dropped). Indeed, since $\bar{u}_t = 0$ for all t in the homogeneous response, one has $\ddot{u}_t = 0$ and there is no inertial effect. However, when the length L of the ring is large enough in comparison with the material characteristic length ℓ, it is possible to construct many other solutions of the quasi-static problem where the damage field α_t localizes in a subdomain of the ring (of size of the order of ℓ) for $t > \varepsilon_c/\dot{\varepsilon}_0$, see (Pham and Marigo 2011) for the method of construction of such solution with damage localization. In other words, a bifurcation occurs at $t = \varepsilon_c/\dot{\varepsilon}_0$, i.e. at the end of the elastic phase. Moreover, one deduces from the stability criterion based on the total energy functional $\mathcal{E}_t$ that the homogeneous response is unstable for $t > \varepsilon_c/\dot{\varepsilon}_0$, see (Pham et al. 2011b) for the details of such a proof. Unfortunately, it is not possible to extend these properties of bifurcation and instability in the dynamical context for the following reasons:

1. *The issue of the uniqueness* Any attempt to find a bifurcation from the homogeneous response at some time t_0 has failed. The main reason is due to the fact that the inertial effects force the velocity field to be continuous in time, what is not required in a quasi-static setting. Even if a complete proof is not available, it seems that the homogeneous response is the unique solution of the dynamical problem (21)–(23). However, as we will show in the next sections, the numerical simulations of this problem show that the homogeneous response disappears after a certain time (which depends on the internal length ℓ and the strain rate $\dot{\varepsilon}_0$) and is followed by a solution with many localizations of the damage field which will give rise finally to the fragmentation of the ring.
2. *The issue of the stability* In dynamics we can no more use the criterion of stability based on the energy $\mathcal{E}_t$ as in quasi-static. We must replace it by another stability criterion. The numerical simulations suggests that the homogeneous is unstable after a certain time in the sense that any small perturbation will grow rapidly. So, we will study how small imperfections (which can be simply due to numerical approximations) can perturb the homogeneous response and we will use that as stability criterion, like in Mercier and Molinari (2003), Rodríguez-Martinez et al. (2013), Vaz-Romero et al. (2017) and Ravi-Chandar and Triantafyllidis (2015).

3.4 Non-dimensionalisation of the problem

3.4.1 Introduction of dimensionless quantities

In order to simplify the numerical treatment of the problem, it is more convenient to introduce dimensionless quantities. The relevant parameters for obtaining the number of fragments will emerge from this non-dimensionalisation procedure. Specifically, we choose the length L of the ring as the reference length, the wave speed $c_0 = \sqrt{E_0/\rho}$ as the reference velocity and hence the time $t_0 = L/c_0$ as the reference time. Therefore, the dimensionless space and time variables are defined as

$$\tilde{x} = \frac{x}{L}, \quad \tilde{t} = \frac{c_0 t}{L}.$$

Moreover, we take σ_c as the reference stress and $\varepsilon_c = \sigma_c/E_0$ as the reference strain. Therefore, the dimensionless displacement, strain, stress and damage fields are defined as follows:

$$\tilde{u}(\tilde{x},\tilde{t}) = \frac{\bar{u}(x,t)}{\varepsilon_c L}, \quad \tilde{\varepsilon}(\tilde{x},\tilde{t}) = \frac{\varepsilon(x,t)}{\varepsilon_c},$$
$$\tilde{\sigma}(\tilde{x},\tilde{t}) = \frac{\sigma(x,t)}{\sigma_c}, \quad \tilde{\alpha}(\tilde{x},\tilde{t}) = \alpha(x,t).$$

After inserting these dimensionless quantities into the problem (22), it turns out that the resulting dimensionless problem depends only on the two following parameters:

$$\tilde{\ell} = \frac{\ell}{L}, \qquad \tilde{\dot{\varepsilon}}_0 = \frac{L\dot{\varepsilon}_0}{c_0\varepsilon_c}. \tag{26}$$

Specifically, one gets

$$\begin{cases} \textit{motion equation}: & \dfrac{\partial\tilde{\sigma}}{\partial\tilde{x}} - \dfrac{\partial^2\tilde{u}}{\partial\tilde{t}^2} = 0 \\ \textit{stress–strain relation}: & \tilde{\sigma} = (1-\tilde{\alpha})^2\left(\dfrac{\partial\tilde{u}}{\partial\tilde{x}} + \tilde{\dot{\varepsilon}}_0\tilde{t}\right) \\ \textit{irreversibility condition}: & \dfrac{\partial\tilde{\alpha}}{\partial\tilde{t}} \geq 0 \\ \textit{damage criterion}: & (1-\tilde{\alpha})\left(\dfrac{\partial\tilde{u}}{\partial\tilde{x}} + \tilde{\dot{\varepsilon}}_0\tilde{t}\right)^2 \\ & \quad + \tilde{\ell}^2\dfrac{\partial^2\tilde{\alpha}}{\partial\tilde{x}^2} \leq 1 \\ \textit{consistency relation}: & \left((1-\tilde{\alpha})\left(\dfrac{\partial\tilde{u}}{\partial\tilde{x}} + \tilde{\dot{\varepsilon}}_0\tilde{t}\right)^2 \right. \\ & \quad \left. + \tilde{\ell}^2\dfrac{\partial^2\tilde{\alpha}}{\partial\tilde{x}^2} - 1\right)\dfrac{\partial\tilde{\alpha}}{\partial\tilde{t}} = 0 \end{cases} \tag{27}$$

Therefore, the response depends only on the ratio $\tilde{\ell}$ between the internal length of the material and the length of the ring, and on the dimensionless rate of expansion $\tilde{\dot{\varepsilon}}_0$ of the ring. In particular the homogeneous damage response when the ring starts from its sound natural state reads now in terms of dimensionless quantities

$$\tilde{\alpha}^*(\tilde{t}) = \begin{cases} 0 & \text{when} \quad 0 \leq \tilde{\dot{\varepsilon}}_0\tilde{t} \leq 1 \\ 1 - \dfrac{1}{\tilde{\dot{\varepsilon}}_0^2\tilde{t}^2} & \text{when} \quad \tilde{\dot{\varepsilon}}_0\tilde{t} > 1 \end{cases} \tag{28}$$

and is, of course, independent of $\tilde{\ell}$.

3.4.2 Practical orders of magnitude of the two dimensionless parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$

The gradient damage model involves in 1D three material parameters: E_0, σ_c and ℓ. The first two can be obtained from the stress–strain response of the volume element. The last one, ℓ, cannot be obtained directly since it appears only in non homogeneous responses of the material. It can be obtained from the measurement of the surface energy parameter G_c or equivalently from the toughness K_c of the material. Indeed, as it is shown in Pham et al. (2011a) and Pham and Marigo (2011), the surface energy necessary to create a crack in a sound material is given by

$$G_c = \frac{4\sqrt{2}}{3}\frac{\sigma_c^2 \ell}{E_0} \tag{29}$$

or equivalently, by $G_c = K_c^2/E_0$, the toughness reads as

$$K_c = \sqrt{\frac{4\sqrt{2}}{3}}\sigma_c\sqrt{\ell} \approx 1.37\sigma_c\sqrt{\ell}. \tag{30}$$

Therefore the measurement of σ_c and K_c allows us to determine ℓ.

Let us consider for example the case of a ring in alumina ceramic (Zhou and Wang 2009; Zinszner et al. 2015) with length $L = 0.1\,\text{m}$, mass density $\rho = 3900\,\text{kg/m}^3$, Young modulus $E_0 = 400\,\text{GPa}$, fracture toughness $K_c = 5\,\text{MPa}\sqrt{\text{m}}$ and ultimate tensile strength $\sigma_c = 400\,\text{MPa}$. Hence, (30) gives

$$\ell = 8\times\ 10^{-5}\,\text{m}, \qquad \tilde{\ell} = 8\times\ 10^{-4}.$$

As far as the rate of expansion is concerned, the wave speed being $c_0 = 1\times 10^4\,\text{m/s}$, the reference time is $t_0 = 1\times\ 10^{-5}\,\text{s}$, whereas the critical strain is $\varepsilon_c = 1\times\ 10^{-3}$. Therefore, if the ring is expanded with a strain rate in the range of 1 to $10^4\,\text{s}^{-1}$, one gets

$$\dot{\varepsilon}_0 \sim 1\text{–}10^4\text{s}^{-1}, \qquad \tilde{\dot{\varepsilon}}_0 \sim 10^{-2}\text{–}100.$$

3.5 Numerical implementation

The displacement and damage fields are discretized in space using the standard Lagrange $P1$ elements. In 1D, the elements of the mesh have the same length Δx and a periodic condition is imposed. For the time discretization, we only consider a constant time step Δt and hence the fields are computed at the instants $\tilde{t}_i = i\Delta t$, $i \in \mathbb{N}$. For the computations, we used the *FeniCS* library (Mardal et al. 2012).

For solving the equation of motion, we have tested two different schemes, the *Explicit Newmark Scheme* and the *Generalized Midpoint Rule Scheme*, which are briefly presented below. It is important to remark that we obtained the same results with both schemes, assuring that the results obtained in the following sections are not a consequence of the discretization used. We will denote by u_h and α_h the displacement and damage fields obtained after the spatial discretization. At the instant $\tilde{t}_i$, we will denote the discrete displacement by u_h^i, the discrete velocity by v_h^i, the discrete acceleration by a_h^i and the discrete damage field by α_h^i.

The damage problem is treated as a constrained optimization problem. We first define the (dimensionless) potential energy at time $\tilde{t}_i$ as

$$\mathcal{E}_i(\tilde{u},\tilde{\alpha}) = \int_0^1 \left(\frac{1}{2}(1-\tilde{\alpha})^2\left(\frac{\partial\tilde{u}}{\partial\tilde{x}} + \tilde{\dot{\varepsilon}}_0\tilde{t}_i\right)^2 + \tilde{\alpha} + \frac{1}{2}\tilde{\ell}^2\left(\frac{\partial\tilde{\alpha}}{\partial\tilde{x}}\right)^2\right)d\tilde{x}. \tag{31}$$

Since the potential energy is convex with respect to the damage field, it is clear that the damage criterion at time $\tilde{t}_i$ for a given displacement field u_h^i is equivalent to minimizing the potential energy at that time under the constraint of irreversibility $\alpha_h^i \geq \alpha_h^{i-1}$. In other words, the discrete damage profile satisfies

$$\alpha_h^i = \arg\min_{\alpha_h^{i-1}\leq\alpha\leq 1} \mathcal{E}_i(u_h^i,\alpha). \tag{32}$$

To solve this constrained minimization problem, the *GPCG* (Moré and Toraldo 1991) algorithm is used.

3.5.1 Explicit newmark scheme

This method is commonly used to solve second order (linear or non-linear) differential equations and we will use it to solve the motion equation. After a spatial discretization using the finite elements method, we obtain the system

$$\mathcal{M}\ddot{u}_h(\tilde{t}) + \mathcal{K}(\alpha_h)u_h(\tilde{t}) = f(\tilde{t}), \tag{33}$$

where $\mathcal{M}$ is the constant mass matrix, $\mathcal{K}$ is the damage dependent stiffness matrix, and the vector $f(\tilde{t})$ contains the applied body forces (due in particular to the prescribed strain rate $\tilde{\dot{\varepsilon}}_0$). Note that in the present context all these quantities are dimensionless.

We will approximate $u_h(\tilde{t})$, $\dot{u}_h(\tilde{t})$ and $\ddot{u}_h(\tilde{t})$ by the sequences u_h^i, v_h^i and a_h^i satisfying

$$\begin{cases} u_h^{i+1} = u_h^i + \Delta t v_h^i + \frac{(\Delta t)^2}{2} a_h^i \\ v_h^{i+1} = v_h^i + \frac{\Delta t}{2}(a_h^i + a_h^{i+1}) \\ \mathcal{M} a_h^{i+1} = f(t^{i+1}) - \mathcal{K}(\alpha_h^{i+1}) u_h^{i+1}. \end{cases} \tag{34}$$

Therefore, after initialization of the algorithm, we use at each instant $\tilde{t}_i$ the following scheme:

(1) Update boundary conditions.
(2) Calculate $u_h^{i+1} = u_h^i + \Delta t v_h^i + \frac{(\Delta t)^2}{2} a_h^i$.
(3) Solve the damage problem to find α_h^{i+1}.
(4) Find the acceleration by solving $\mathcal{M} a_h^{i+1} = f(\alpha_h^{i+1}) - \mathcal{K}(\alpha_h^{i+1}) u_h^{i+1}$.
(5) Find the velocity $v_h^{i+1} = v_h^i + \frac{\Delta t}{2}(a_h^i + a_h^{i+1})$.
(6) Advance to time step $i + 1$.

This explicit Newmark scheme is stable provided that $\Delta t \leq \Delta x$. To prevent possible instabilities caused by the damage process, we take $\Delta t = \Delta x/2$.

3.5.2 Generalized midpoint rule scheme

In this section, we detail only the unidimensional algorithm, even though its extension to the multidimensional should not pose any problem. The algorithm consists of a loop during a time step where one alternatively solves the equation of motion at given damage, then computes the damage field at given strain field, and iterates until convergence of both variables before passing to the next time step. We follow the ideas presented in Simo and Hughes (1998) where it is proposed to fix $\theta \in [0, 1]$ and, define u_θ as the weighted average between the values of u_h^i and u_h^{i+1}:

$$u_\theta = (1 - \theta) u_h^i + \theta u_h^{i+1}. \tag{35}$$

The same notation will be used for the other variables.

We have the following implicit system of equations for an elastic material:

$$\begin{cases} u_h^{i+1} - u_h^i = \Delta t v_\theta \\ \displaystyle\int_0^1 \rho \frac{(v_h^{i+1} - v_h^i)}{\Delta t} w d\tilde{x} + \int_0^1 \sigma_\theta w' \, d\tilde{x} = 0, \forall w. \end{cases} \tag{36}$$

This scheme is stable and non-dissipative if $\theta = 1/2$ (Simo and Hughes 1998). In order to take damage evolution into account, we first have to define at which instant between $\tilde{t}_i$ and $\tilde{t}_{i+1}$ how the damage criterion is verified. There is also a question of how to define σ_θ. For instance, we can take

$$\begin{cases} \sigma_\theta^1 &= (1 - \theta)\sigma_i + \theta \sigma_h^{i+1} \\ \sigma_\theta^2 &= E(\alpha_\theta) u'_\theta \\ \sigma_\theta^3 &= \big((1 - \theta) E(\alpha_h^i) + \theta E(\alpha_h^{i+1})\big) u'_\theta. \end{cases} \tag{37}$$

Numerical experiments have shown that, whatever is the chosen option, the system will have the same behavior when the time-step and mesh size are small enough. Finally, at each instant $\tilde{t}_i$, we run the following algorithm:

(1) Solve the displacement problem:

$$\begin{cases} u_h^{i+1} - u_h^i = \Delta t v_{1/2} \\ \displaystyle\int_0^1 \rho \frac{(v_h^{i+1} - v_h^i)}{\Delta t} w \, d\tilde{x} + \int_0^1 \sigma_{1/2} w' \, d\tilde{x} = 0, \forall w \end{cases} \tag{38}$$

(2) Solve the damage problem using the displacement u_h^{i+1} to find α_h^{i+1}
(3) Verify the convergence of u_h^{i+1} and α_h^{i+1}. Repeat if it does not converge, otherwise go to the next time step.

4 Calculation of the fragmentation of the 1D ring

In this section, we first present the numerical results that we obtained using the algorithms described in the previous section for different values of the dimensionless structural parameters $(\tilde{\ell}, \tilde{\dot{\varepsilon}}_0)$ and of the numerical parameter Δx (recalling that we take $\Delta t = \Delta x/2$ for the Newmark scheme). It turns out that the numerical solution corresponds to the uniform response until a critical time (which depends on the parameters). Then the damage field rapidly localizes to form several fragments. We discuss in particular the dependence of the number of fragments on all the parameters. In a second time, we try to explain those numerical results by using a *Linear Stability Analysis* of the homogeneous response where we study the growth in time of initial small perturbations.

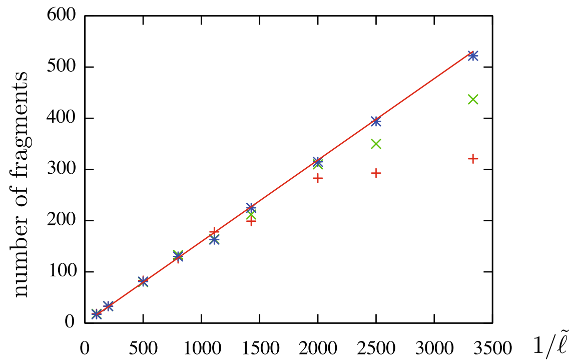

Fig. 1 Number of computed fragments versus $1/\tilde{\ell}$ when $\tilde{\dot{\varepsilon}}_0 = 0.5$ and for three mesh sizes ($\Delta x = 5\times10^{-4}$, 2×10^{-4}, 1×10^{-4}). The red line corresponds to the converged value

4.1 Numerical simulations

The results are sensitive to the numerical parameters (mesh size, convergence threshold, ...). In particular the mesh size must be fine enough so that the number of fragments does not change. The convergence of the number of fragments when Δx goes to 0 has to be verified whereas the convergence of the displacement or the damage fields is more debatable. Indeed, since the problem is essentially invariant by translation because of the initial and periodic conditions, one cannot require that the fragments are located at given points but only that their number et their relative location have an intrinsic character. Accordingly, for each set of the parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$, we run several simulations with different values of Δx and check if the number of fragments converge when $\Delta x \to 0$. The Fig. 1 shows that, when $\tilde{\dot{\varepsilon}}_0 = 0.5$, the smaller the parameter $\tilde{\ell}$, the greater the number of elements in order that the computed number of fragments does not change. Specifically, one can see that for $1/\tilde{\ell} \leq 10^3$, we obtain the same results using $2\ 10^3$, $5\ 10^3$ or 10^4 elements. For $1/\tilde{\ell} \leq 2\ 10^3$, $5\,10^3$ elements are sufficient to obtain the converged number of cracks. Finally, we consider that the simulation results are accurate if the mesh size Δx is such that $\Delta x \leq \tilde{\ell}/3$. We have checked that this condition is valid for other values of $\tilde{\dot{\varepsilon}}_0$.

Typical evolutions of the damage fields are shown in Subfigures (1)–(4) of Fig. 2 for four sets of the parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$. In each case, one remarks that the response is practically homogeneous and follows the damage evolution given by (28) during a time interval which depends on $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$. The absence of a perfect uniformity of the damage fields is due to small perturbations introduced by numerical approximations. For instance, when $\tilde{\ell} = 0.01$ and $\tilde{\dot{\varepsilon}}_0 = 1$, one can see on Subfigure (5) of Fig. 2 that at time $\tilde{t} = 1.1$ the computed damage field oscillates around the value $\alpha = 0.1743$ with an amplitude of 3×10^{-9}. (That computed value $\alpha = 0.1743$ is slightly higher than that given by (28), $\alpha^* = 0.1736$.) Thus, those small oscillations progressively grow with time until a critical time. After that critical time the amplitude of the oscillations become finite, the damage localizes in several zones and it grows rapidly at the center of these zones to finally reach the value 1 which corresponds to a crack. Practically all the cracks nucleate at the same time and their spatial distribution is practically periodic. After this nucleation time, the damage field does not evolve practically any more. Accordingly, one can distinguish the two following characteristics of the simulated damage evolution:

1. *The critical time after which the response is really no more homogeneous.* That critical time and consequently the value of the damage from which localizations occur increases rapidly with the strain rate $\tilde{\dot{\varepsilon}}_0$ at given internal length $\tilde{\ell}$. Its dependence on $\tilde{\ell}$ at given $\tilde{\dot{\varepsilon}}_0$ is also strong as one can see by comparing the results in the last two sets of parameters in Fig. 2.
2. *The number of fragments.* This number seems essentially piloted by the internal length and hence by the parameter $\tilde{\ell}$. On Fig. 1, once can see that the number of fragments increases linearly with $1/\tilde{\ell}$ for $\tilde{\dot{\varepsilon}}_0 = 0.5$. This linear behavior holds true for every other value of $\tilde{\dot{\varepsilon}}_0$ tested between 10^{-4} and 10^2, as long as the mesh is fine enough. The dependence on the strain rate and hence on $\tilde{\dot{\varepsilon}}_0$ seems more marginal (even if one observes a tendency to an increase of the number of fragments with the strain rate) as long as the product $\tilde{\ell}\tilde{\dot{\varepsilon}}_0$ remains small with respect to 1. When $\tilde{\dot{\varepsilon}}_0$ is too large so that $\tilde{\ell}\tilde{\dot{\varepsilon}}_0$ is no more small, since the departure of the homogeneous response occurs when the damage is close to 1, the concept of fragments becomes meaningless because practically all the body is broken. Moreover, the accuracy of the numerical results is in that case less good. At the other end, when the strain rate $\tilde{\dot{\varepsilon}}_0$ is small, the localization appears practically as soon as the damage starts and the number of fragments is more sensitive to any imperfection as one can see on Fig. 5.

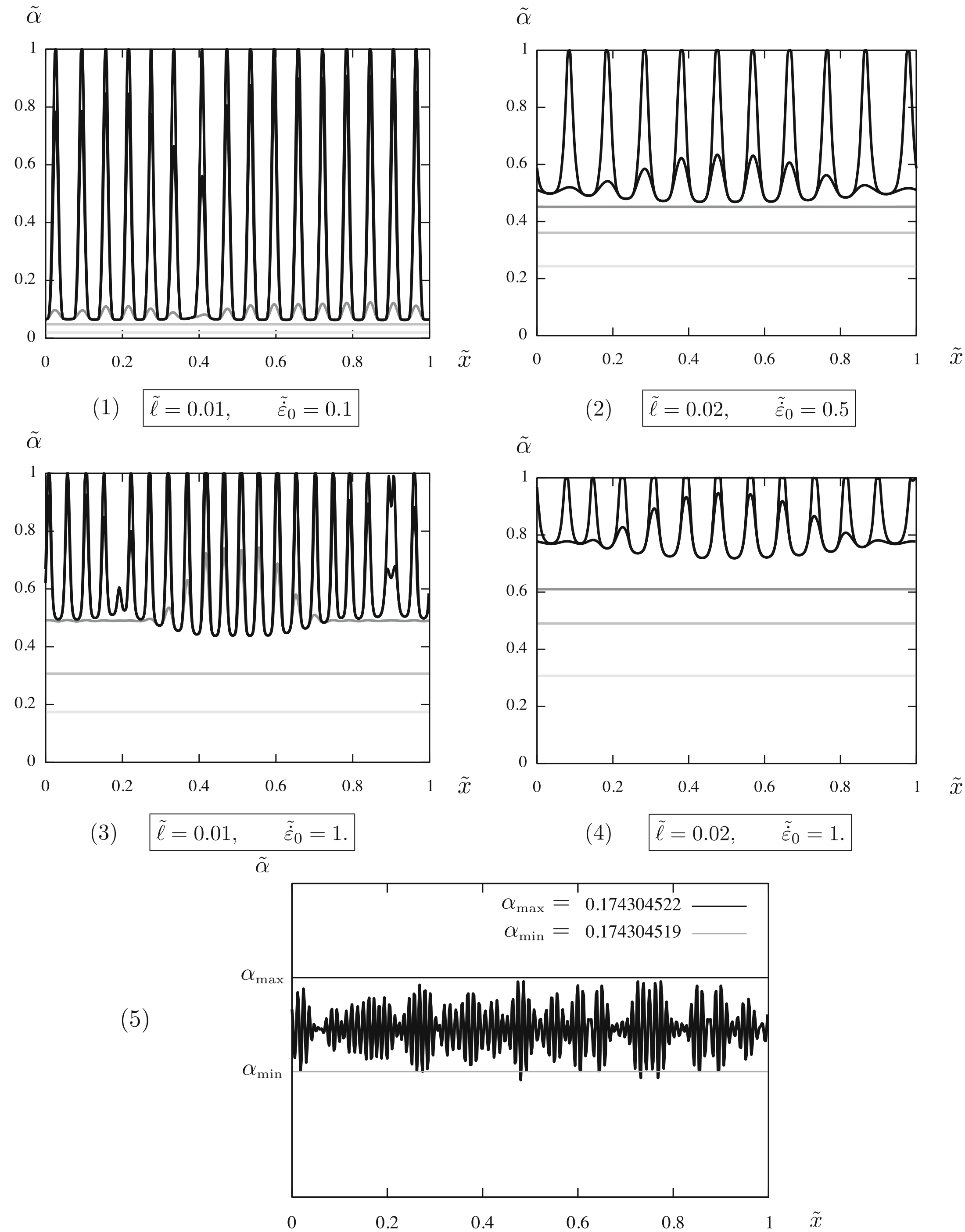

Fig. 2 (1)–(4) Graphs of the damage field $\tilde{x} \mapsto \tilde{\alpha}(\tilde{x}, \tilde{t})$ at five different times $\tilde{t}$ for four sets of the parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$. In each case, the response is practically homogeneous up to a critical time which depends on the parameters. At the final time, a certain number of cracks (points $\tilde{x}$ where $\tilde{\alpha} = 1$) have been created, a zone between two cracks corresponds to a fragment; (5) graph of the damage field $\tilde{x} \mapsto \tilde{\alpha}(\tilde{x}, \tilde{t})$ at time $\tilde{t} = 1.1$ when $\tilde{\ell} = 0.01$ and $\tilde{\dot{\varepsilon}}_0 = 1$. The damage is practically uniform with small oscillations of amplitude 3×10^{-9}

4.2 Linear analysis of stability of the homogeneous response

This subsection is devoted to an analysis of stability of the homogeneous response. That issue of stability in dynamics has been addressed in many works, see for instance (Mercier and Molinari 2003; Rodríguez-Martinez et al. 2013; Vaz-Romero et al. 2017; Ravi-Chandar and Triantafyllidis 2015) where similar non linear dynamical problems have been considered. However our stability analysis differs from the usual one in the sense that we consider here the full (linearized) problem of evolution of the imperfections in time whereas in general the coefficients of the equation of perturbation are frozen at a given time.

4.2.1 The linearized equation of perturbation

Let us analysis the growth of a small perturbation from the homogeneous response, $\tilde{u} = 0$, $\tilde{\alpha}(\tilde{x}, \tilde{t}) = \tilde{\alpha}^*(\tilde{t})$ with $\tilde{\alpha}^*$ given by (28). Since the response is purely elastic until the time $\tilde{t}_0 = 1/\dot{\tilde{\varepsilon}}_0$, one can starts the analysis from $\tilde{t}_0$. Assuming that a small perturbation has be introduced in the response (the origin of which can be purely numeric) at $\tilde{t}_0$, let us study its evolution in time. Denoting by $\delta\tilde{u}$, $\delta\tilde{\varepsilon}$, $\delta\tilde{\sigma}$ and $\delta\tilde{\alpha}$ the perturbation of the displacement, the strain, the stress and the damage fields, one obtains after linearization of the set of equations (26) around the homogeneous response that the disturbed fields must satisfy:

$$\begin{cases} \delta\tilde{\sigma}' = \delta\ddot{\tilde{u}} \\ \delta\tilde{u}' = \delta\tilde{\varepsilon} \\ \delta\tilde{\sigma} = \dfrac{1}{(\dot{\tilde{\varepsilon}}_0\tilde{t})^4}\delta\tilde{\varepsilon} - \dfrac{2}{\dot{\tilde{\varepsilon}}_0\tilde{t}}\delta\tilde{\alpha} \\ (\dot{\tilde{\varepsilon}}_0\tilde{t})^2\delta\tilde{\alpha} - \tilde{\ell}^2\delta\tilde{\alpha}'' = \dfrac{2}{\dot{\tilde{\varepsilon}}_0\tilde{t}}\delta\tilde{\varepsilon} \end{cases} \tag{39}$$

where the prime and the dot indicate partial derivatives with respect to $\tilde{x}$ and $\tilde{t}$ respectively. The last equation in (39) is deduced from the consistency relation owing to the perturbation is assumed small and $d\tilde{\alpha}^*/d\tilde{t} > 0$. Differentiating with respect to $\tilde{x}$ the first equation of (39) and combining with the other three, one can eliminate $\delta\tilde{u}$, $\delta\tilde{\sigma}$ and $\delta\tilde{\alpha}$ to finally obtain the following partial differential equation of the fourth order for $\delta\tilde{\varepsilon}$:

$$\tilde{\ell}^2\delta\tilde{\varepsilon}'''' - \tilde{\ell}^2(\dot{\tilde{\varepsilon}}_0\tilde{t})^4\delta\ddot{\tilde{\varepsilon}}'' + 3(\dot{\tilde{\varepsilon}}_0\tilde{t})^2\delta\tilde{\varepsilon}'' + (\dot{\tilde{\varepsilon}}_0\tilde{t})^6\delta\ddot{\tilde{\varepsilon}} = 0. \tag{40}$$

Let us note that the fourth order terms (which are due to the regularization of the damage model) correspond to an hyperbolic equation. But the second order terms (which are the unique terms when the model is not regularized) correspond to an elliptic equation, consequence of the softening character of the model.

Let us now consider a perturbation of the form

$$\delta\tilde{\varepsilon}(\tilde{x}, \tilde{t}) = \sin(2\pi N\tilde{x})\phi(\tilde{t}) \tag{41}$$

where $N \in \mathbb{N}^*$ represents the given perturbation number and $\phi(\tilde{t})$ is the amplitude of that mode of perturbation at time $\tilde{t}$. Inserting (41) into (40) gives the following second order linear differential equation governing the evolution of the amplitude of the perturbation:

$$\frac{d^2\phi}{d\tilde{t}^2}(\tilde{t}) = \frac{4\pi^2N^2}{(\dot{\tilde{\varepsilon}}_0\tilde{t})^4}\,\frac{3(\dot{\tilde{\varepsilon}}_0\tilde{t})^2 - 4\pi^2N^2\tilde{\ell}^2}{(\dot{\tilde{\varepsilon}}_0\tilde{t})^2 + 4\pi^2N^2\tilde{\ell}^2}\,\phi(\tilde{t}). \tag{42}$$

4.2.2 Analytic approximation

This subsection is devoted to the analytic study of the differential equation (42). Its goal is to obtain an accurate analytical approximation of its solution for the practical range of the parameters $\tilde{\ell}$ and $\dot{\tilde{\varepsilon}}_0$. Let us first make a change of the time variable by setting

$$\tau = \frac{\dot{\tilde{\varepsilon}}_0\tilde{t}}{2\pi N\tilde{\ell}}, \qquad \varphi(\tau) = \phi(\tilde{t}), \tag{43}$$

φ representing the amplitude of the perturbation as a function of τ. Note that the new time variable τ depends on the perturbation number N. Inserting (43) into (42) leads to the following rescaled differential equation for the amplitude φ:

$$\eta^2\frac{d^2\varphi}{d\tau^2}(\tau) = f(\tau)\varphi(\tau) \tag{44}$$

where

$$\eta = \tilde{\ell}\dot{\tilde{\varepsilon}}_0 = \frac{\ell\dot{\varepsilon}_0}{c_0\varepsilon_c} \tag{45}$$

and f is the function of τ defined by

$$f(\tau) = \frac{3\tau^2 - 1}{\tau^4(\tau^2 + 1)}. \tag{46}$$

In practice, if one considers the usual orders of magnitude of $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$ (see Sect. 3.4.2), the parameter η is small. It becomes of the order of 1 only for very short ring or very high strain rate. Consequently, the presence of this small parameter allows us to construct an approximate solution of the differential equation by using classical asymptotic methods. This asymptotic approximation is derived in Appendix A. One obtains that the general solution φ of (44) can be approximated by

$$\varphi(\tau) = \begin{cases} \left(2a_1 \sin\left(\frac{F(\tau)}{\eta} + \frac{\pi}{4}\right) + b_1 \cos\left(\frac{F(\tau)}{\eta} + \frac{\pi}{4}\right)\right) |f(\tau)|^{-1/4} & \text{when} \quad 0 < \tau < \frac{1}{\sqrt{3}} - O(\eta^{2/3}) \\ \left(a_1 \exp\left(-\frac{F(\tau)}{\eta}\right) + b_1 \exp\left(\frac{F(\tau)}{\eta}\right)\right) f(\tau)^{-1/4} & \text{when} \quad \tau > \frac{1}{\sqrt{3}} + O(\eta^{2/3}) \end{cases}, \tag{47}$$

where a_1 and b_1 are arbitrary constants and $F(\tau)$ is given by

$$F(\tau) = \left| \int_{1/\sqrt{3}}^{\tau} \sqrt{|f(s)|} ds \right|. \tag{48}$$

In the neighborhood of $1/\sqrt{3}$, there exists a boundary layer which matches the two asymptotic approximations given in (47), but it is not reproduced here, see Appendix A. The two fundamentals solutions, the first one corresponding to $(a_1, a_2) = (1, 0)$ and the second one to $(a_1, a_2) = (0, 1)$, are plotted on Fig. 3 for $\eta = 0.3$. One sees that the first one oscillates before decreasing and, hence, never takes large values. On the contrary, the second one oscillates when $\tau < 1/\sqrt{3}$ before growing exponentially when $\tau > 1/\sqrt{3}$. Those properties of oscillation with or without a rapid growth are true for any small enough value of η, and even, the smaller is η, the more accentuated are those properties. That means that one can disregard the first fundamental solution because it cannot generate an instability. Moreover, the instability generated by the second fundamental solution can only occur when $\tau > 1/\sqrt{3}$. Therefore, in the study of the instability by growth of the perturbations, one can consider that the evolution of the amplitude is simply given by

$$\varphi(\tau) = \exp\left(\frac{F(\tau)}{\eta}\right) f(\tau)^{-1/4} \qquad \text{for} \quad \tau > \frac{1 + C_\eta}{\sqrt{3}}, \tag{49}$$

where C_η tends to 0 like $\eta^{2/3}$ when η tends to 0. In (49) the constant b_1 can be taken equal to 1 because the problem is linear and because we are simply interested on the relative growth of the imperfections.

4.2.3 Determination of the perturbation number governing the instability

After the end of the elastic phase (i.e. from $\tilde{t} = 1/\tilde{\dot{\varepsilon}}_0$), the evolution of the perturbations is governed by (44). That corresponds, for the perturbation number N, to the rescaled time $\tau_0(N)$ given by

$$\tau_0(N) = \frac{1}{2\pi N \tilde{\ell}}. \tag{50}$$

One considers that the mode of instability will correspond to the perturbation number N^* which is the first whose amplitude is amplified by a given factor S. Specifically, for a given perturbation number N, let us denote by $\tau_1(N)$ the rescaled time such that

$$\varphi(\tau_1(N)) = S\varphi(\tau_0(N)). \tag{51}$$

By virtue of (43) and (50), the associated dimensionless time $\tilde{t}_1(N)$ is given by

$$\tilde{t}_1(N) = \frac{\tau_1(N)}{\tau_0(N)\tilde{\dot{\varepsilon}}_0}.$$

Therefore, the problem consists in finding N^* which minimizes $\tau_1(N)N$ over all $N \in \mathbb{N}$.

From a practical point of view, when using the library *FEniCS*, we find that $\max(\varepsilon) - \min(\varepsilon) \approx 10^{-16}$ in the first iteration. Accordingly, if one considers that a perturbation becomes visible when its amplitude is greater than 10^{-3}, we take $S = 10^{13}$. We will also study the influence of that choice of S on the results.

To find the mode of instability one can disregard all the perturbation numbers N such that $\tau_0(N) \leq 1/\sqrt{3}$ because their amplitude will first oscillate and will grow rapidly only when τ will be larger than $1/\sqrt{3} + O(\eta^{2/3})$. Therefore the ratio $\tau_1(N)/\tau_0(N)$

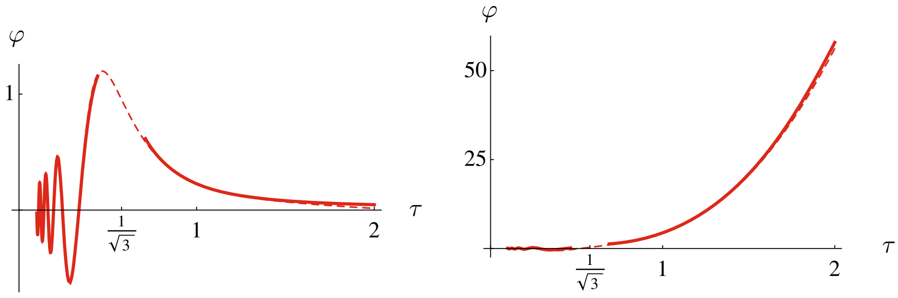

Fig. 3 The two fundamental solutions of the differential equation (44) when $\eta = 0.3$. The dashed lines correspond to the exact solutions of (44) computed with a classical Runge–Kutta method, the plain lines to their approximation by (47) (outside the boundary layer near $1/\sqrt{3}$). Note that the approximation is very good even though η is not very small

for such perturbation number will be greater than the ratio of the perturbation number N such that $\tau_0(N) \geq 1/\sqrt{3} + O(\eta^{2/3})$. Accordingly, one can solve (51) by using the expression (49) for φ. Specifically, taking the logarithm of (51) and using (49), one gets that, for a given $\tau_0 > (1 + C_\eta)/\sqrt{3}$, τ_1 must satisfy:

$$F(\tau_1) - F(\tau_0) - \frac{\eta}{4} \ln \frac{f(\tau_1)}{f(\tau_0)} = \eta \ln S. \tag{52}$$

Since φ is strictly increasing in the region concerned and since η is assumed small, τ_1 is close to τ_0 and we can search τ_1 under the form of the following asymptotic expansion:

$$\tau_1 = \tau_0 + \eta T_1 + \eta^2 T_2 + \cdots . \tag{53}$$

We will only determine the first two terms T_1 and T_2 because we have expanded $\ln \varphi$ only up to the second order, but we could determine the next terms by using the next terms of the expansion of $\ln \varphi$, see "Appendix A". Inserting (53) into (52), expanding $F(\tau_1) - F(\tau_0)$ up to second order and $\ln(f(\tau_1)/f(\tau_0))$ up to first order, one finally gets

$$T_1 = \frac{\ln S}{\sqrt{f(\tau_0)}} \qquad T_2 = -\frac{\ln S(\ln S - 1)\dot{f}(\tau_0)}{4 f(\tau_0)^2}. \tag{54}$$

Hence the time $\tilde{t}_1$ at which the the perturbation number N associated to τ_0 reaches the threshold S is given by

$$\dot{\tilde{\varepsilon}}_0 \tilde{t}_1(\tau_0) = 1 + \eta \frac{\ln S}{\tau_0 \sqrt{f(\tau_0)}} - \eta^2 \frac{\ln S(\ln S - 1)\dot{f}(\tau_0)}{4\tau_0 f(\tau_0)^2}. \tag{55}$$

So, the problem is to find τ_0 which minimizes $\tilde{t}_1(\tau_0)$. This optimal τ_0 depends of course on η and its dependence can still be expanded in powers of η:

$$\tau_0 = \tau_{00} + \eta \tau_{01} + \cdots . \tag{56}$$

We will only determine the first two terms τ_{00} and τ_{01} (the next ones requiring that one determines the next terms of the expansion of $\ln \varphi$). By virtue of (55), the leading term is $\tau_{00} > 1/\sqrt{3}$ which maximizes $\tau^2 f(\tau)$ over all $\tau > 1/\sqrt{3}$. Using (46), one gets

$$\tau^2 f(\tau) = \frac{3\tau^2 - 1}{\tau^2(\tau^2 + 1)} = \frac{4}{\tau^2 + 1} - \frac{1}{\tau^2},$$

from which one easily deduces that $\tau_{00} = 1$. Then taking $\tau_0 = 1 + \eta \tau_{01} + \cdots$ and expanding $\tilde{t}_1(\tau_0)$ up to the third order, one gets

$$\begin{aligned} \dot{\tilde{\varepsilon}}_0 \tilde{t}_1(\tau_0) = {} & 1 + \eta \ln S + \frac{\eta^2}{2} \ln S(\ln S - 1) \\ & + \eta^3 \left(\ln S\, \tau_{01}^2 + \ln S(\ln S - 1)\tau_{01} + C_3 \right) \\ & + O(\eta^4). \end{aligned} \tag{57}$$

In (57) the constant C_3 contains extra terms than those coming from (55). In fact C_3 involves the first three terms of the expansion of $\ln \varphi$ (called z_0, z_1 and z_2 in the Appendix A) calculated at $\tau = 1$. In particular C_3 involves $\dot{z}_2(1)$ which can be obtained from z_0, z_1 and their derivatives at $\tau = 1$. But since C_3 does not depend on τ_{01}, it is not necessary to calculate it. Finally, minimizing (57) with respect to τ_{01} gives

$$\tau_{01} = -\frac{\ln S - 1}{2}.$$

In conclusion, the mode of instability corresponds to the perturbation number N^* and is given by

$$N^* = \frac{1}{\pi \tilde{\ell}\Big(2 - (\ln S - 1)\tilde{\ell}\dot{\tilde{\varepsilon}}_0\Big)}. \qquad (58)$$

The time at which the instability occurs is

$$\tilde{t}_1 = \frac{1}{\dot{\tilde{\varepsilon}}_0}\left(1 + \tilde{\ell}\dot{\tilde{\varepsilon}}_0 \ln S + \frac{\tilde{\ell}^2\dot{\tilde{\varepsilon}}_0^2}{2}\ln S(\ln S - 1)\right) \qquad (59)$$

and hence the value α_1 of the damage at which the homogeneous response becomes unstable is given by

$$\begin{aligned}\alpha_1 &= 1 - \frac{1}{\dot{\tilde{\varepsilon}}_0^2\tilde{t}_1^2}\\ &= 1 - \frac{1}{\left(1 + \tilde{\ell}\dot{\tilde{\varepsilon}}_0 \ln S + \frac{\tilde{\ell}^2\dot{\tilde{\varepsilon}}_0^2}{2}\ln S(\ln S - 1)\right)^2}.\end{aligned} \qquad (60)$$

To obtain (57)–(60), one uses (28), (45), (50) and the fact that $\tau_0 = 1$, $f(1) = 1$, $\dot{f}(1) = -2$ and $\ddot{f}(1) = 2$.

Remark 1 Note that the mode of instability depends at the first order only on the dimensionless $\tilde{\ell}$, but neither on the expansion strain rate $\dot{\tilde{\varepsilon}}_0$ nor on the threshold S. But the time and the damage value at which the instability occurs depends on all the three parameters $\tilde{\ell}$, $\dot{\tilde{\varepsilon}}_0$ and S. However this dependence on S is rather small as we will show in the next section. Let us recall that all these properties have been obtained by assuming that the dimensionless parameter $\tilde{\ell}\dot{\tilde{\varepsilon}}_0$ is small (by comparison to 1) and varies in a range of values which are relevant for practical expansion tests on ceramic like materials. All the expressions above for N^* and α_1 depend only on the function f given by (46) which itself is directly related to the damage model (18). All the other models that we have tested give the same qualitative results for the dependence of N^* and α_1 on $\tilde{\ell}$ and $\dot{\tilde{\varepsilon}}_0$, only the coefficients change in the expressions (58)–(60).

4.3 Numerical comparisons

We compare here the results obtained from the direct simulations and those coming from the Stability Analysis.

1. *Number of cracks.* If one compares the number of cracks given by the simulation and the perturbation number N^* corresponding to the mode of instability (Eq. (58)), the agreement is very good for all the range of values of the parameters $\tilde{\ell}$ that we have tested provided that $\dot{\tilde{\varepsilon}}_0$ is neither too small nor too large. The Fig. 4 and the associated table show the values obtained by the two approaches when $\dot{\tilde{\varepsilon}}_0 = 1$ and $\tilde{\ell}$ varying from 2×10^{-4} to 10^{-2}. The values for N^* are obtained with a threshold $S = 10^{13}$. The numerical simulations show a small dependence of the number of fragments on the strain rate. This small dependence on $\dot{\tilde{\varepsilon}}_0$ is rather well represented by N^* (given by (58)) when $\dot{\tilde{\varepsilon}}_0$ is sufficiently small so that $\tilde{\ell}\dot{\tilde{\varepsilon}}_0$ be small, as one can see on Fig. 5.
2. *Departure from the homogeneous response.* The critical value of α beyond which the response is no more homogeneous is strongly dependent on both parameters $\tilde{\ell}$ and $\dot{\tilde{\varepsilon}}_0$. Moreover, the value α_1 predicted by the Linear Analysis of Stability (Eq. (60)) depends also on the choice of the threshold S. It turns out that this latter dependence on S is rather small, as one can see on Fig. 6. Finally, there is a very good agreement between the computed values of α_1 and those given by (60), as shown in Fig. 6.

In conclusion, one can say that the number of fragments as well as the critical time at which they appear are correctly predicted by the Linear Analysis of Stability. This means that the fragmentation is essentially governed by the growth of imperfections, the number of fragments corresponding to the most unfavorable mode, that is, the mode which grow the fastest.

4.4 Comparison with simplified approaches

In Grady (1982) Grady proposes a simplified approach to estimate the number of fragments. Based on energetic arguments, a 3D analysis leads to a law where the number of fragments N grows with the strain rate $\dot{\varepsilon}_0$ like $\dot{\varepsilon}_0^{2/3}$. However this power law is obtained after neglecting the strain energy stored in the body before its fragmentation. Consequently, an improvement of the approach was developed in Glenn and Chudnovsky (1986) where the stored energy is taken into account. To simplify our presentation, we recall here below the main assumptions and the main steps of the analysis made in Glenn and Chudnovsky (1986) by adapting them to our 1D context of a ring expansion.

$\tilde{\ell}$	Computed number of cracks	N^*
10^{-2}	20	19
$5{\cdot}10^{-3}$	36	34
$2{\cdot}10^{-3}$	81	82
10^{-3}	162	162
$9{\cdot}10^{-4}$	164	179
$7{\cdot}10^{-4}$	237	230
$5{\cdot}10^{-4}$	318	320
$4{\cdot}10^{-4}$	401	400
$3{\cdot}10^{-4}$	534	532
$2{\cdot}10^{-4}$	779	798

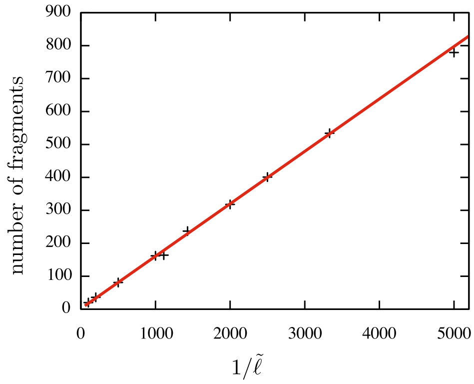

Fig. 4 Comparison between the number of cracks found in the simulations and the mode of instability N^* predicted by the Linear Stability Analysis when $\tilde{\dot{\varepsilon}}_0 = 1$ for different values of $\tilde{\ell}$. In the figure, the plain line corresponds to the number N^* predicted by (58), the marks correspond to the computed values by a direct simulation

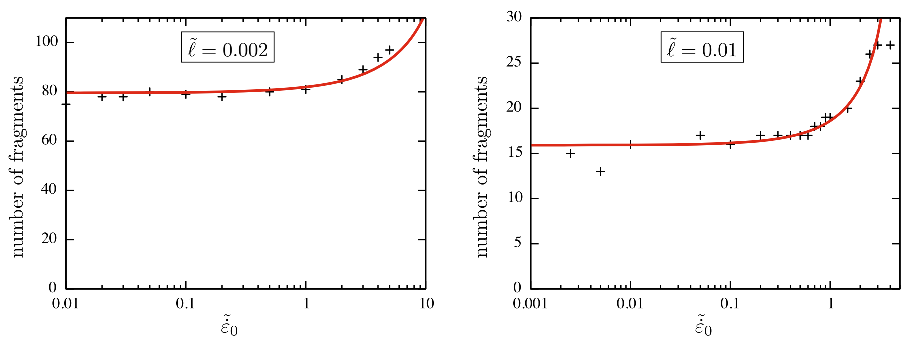

Fig. 5 Number of fragments versus $\tilde{\dot{\varepsilon}}_0$ (in a logarithmic scale) when $\tilde{\ell} = 10^{-2}$ (left) or $\tilde{\ell} = 2 \times 10^{-3}$ (right). The marks correspond to the number of fragments computed by a direct simulation, the plain line to the number of fragments predicted by the linear stability analysis (58) with $S = 10^{13}$

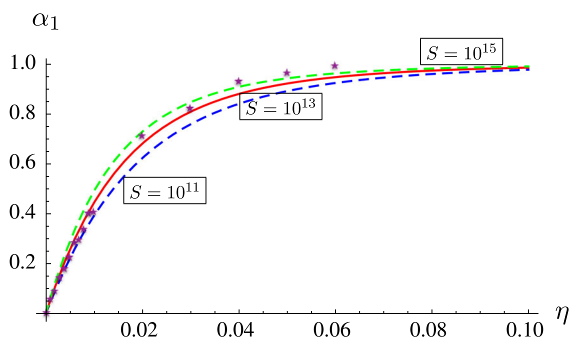

Fig. 6 Value of the damage when the homogeneous response becomes unstable as a function of the dimensionless parameter $\eta = \tilde{\ell}\tilde{\dot{\varepsilon}}_0$. The three curves correspond to the values of α_1 given by (60) for three values of the threshold (the higher the threshold, the higher the curve). The stars correspond to the computed values by using the algorithm presented in Sect. 3.5 with $\tilde{\ell} = 0.01$

4.4.1 *Energy estimates before fragmentation of the ring*

At any time before fragmentation, the velocity field in the ring is $\dot{u}(x) = \dot{\varepsilon}_0 x$ and hence the kinetic energy $\mathcal{K}^-$ is given by

$$\mathcal{K}^- = \int_0^L \frac{1}{2}\rho S \dot{u}(x)^2 dx = \frac{1}{6}\rho S L^3 \dot{\varepsilon}_0^2, \tag{61}$$

where S denotes the area of the cross-section of the ring. At time t before fragmentation, the strain and stress fields are uniform and respectively equal to $\dot{\varepsilon}_0 t$ and $E_0\dot{\varepsilon}_0 t$. Accordingly, assuming that the ring has a linear elastic behavior, the elastic energy stored in the ring at that time is

$$\mathcal{U}(t) = \frac{1}{2}E_0 S L^2 \dot{\varepsilon}_0^2 t^2.$$

Following Glenn and Chudnovsky (1986), one assumes that the fragmentation of the ring occurs at the time t_c when the stress reaches the critical value σ_c (or equivalently when the strain reaches the critical value $\varepsilon_c = \sigma_c/E_0$). Therefore, $t_c = \varepsilon_c/\dot{\varepsilon}_0$ and just before that critical time the elastic energy is given by

$$\mathcal{U}^- = \frac{1}{2}E_0 S L^2 \varepsilon_c^2. \tag{62}$$

4.4.2 *Energy estimates after fragmentation of the ring*

Still like in Glenn and Chudnovsky (1986), one assumes that for $t > t_c$ the ring is separated into N fragments of the same size. So, for $1 \le i \le N$, the ith fragment corresponds to the interval

$$I_i = \left(x_i - \frac{L}{2N}, x_i + \frac{L}{2N}\right) \quad \text{with} \quad x_i = \frac{(2i-1)L}{2N}.$$

Since the fragments are no more submitted to any exterior forces, one assumes that each one will have a rigid motion with the velocity of their mass center x_i. Accordingly, by conservation of the momentum of each fragment, the velocity field will read for $t > t_c$ as

$$\dot{u}(x) = \dot{\varepsilon}_0 x_i = \frac{(2i-1)L}{2N}\dot{\varepsilon}_0, \quad \forall x \in I_i, \quad \forall i, 1 \le i \le N.$$

One deduces that the kinetic energy after fragmentation is given by

$$\begin{aligned}\mathcal{K}^+ &= \sum_{i=1}^{N} \frac{1}{2}\rho S \frac{(2i-1)^2\dot{\varepsilon}_0^2 L^3}{4N^3} \\ &= \frac{1}{6}\rho S L^3 \dot{\varepsilon}_0^2 - \frac{1}{24}\rho S \frac{L^3}{N^2}\dot{\varepsilon}_0^2. \end{aligned} \tag{63}$$

The comparison with (61) leads to

$$\mathcal{K}^+ = K^- - \frac{1}{24}\rho S \frac{L^3}{N^2}\dot{\varepsilon}_0^2. \tag{64}$$

Therefore, the lost of kinetic energy after fragmentation is inversely proportional to N^2 (that expression is also obtained by Miller et al. (1999) in a similar context of 1D fragmentation). In the same time, all the fragments become unstrained and hence the ring looses all its elastic energy so that

$$\mathcal{U}^+ = 0. \tag{65}$$

On the other hand, the fragmented ring gains now a fracture energy and, if one follows Griffith's assumption like in Glenn and Chudnovsky (1986), that energy is proportional to the surface of the created cracks. Therefore, the fracture energy after fragmentation reads as

$$\Gamma = N G_c S, \tag{66}$$

where G_c denotes the Griffith surface energy density.

4.4.3 The resulting equation that governs the number of fragments

By conservation of the energy just before and just after the fragmentation, one gets $\mathcal{K}^- + \mathcal{U}^- = \mathcal{K}^+ + \Gamma$. Using (62), (64) and (66) leads to the following equation which governs the number of fragments:

$$\frac{\sigma_c^2 L}{2E_0} + \frac{\rho L^3 \dot{\varepsilon}_0^2}{24N^2} = G_c N. \tag{67}$$

Equation (67) is essentially the same equation as the one obtained by Glenn and Chudnovsky (1986). It differs only by some coefficients because the analysis in Glenn and Chudnovsky (1986) was made in a 3D context. The Eq. (67) can be rewritten by introducing the material toughness K_c and the speed c_0 of the elastic waves in the sound material:

$$K_c = \sqrt{G_c E_0}, \qquad c_0 = \sqrt{\frac{E_0}{\rho}}.$$

Moreover since the ratio between the toughness K_c and the critical stress σ_c involves a characteristic length, let us introduce ℓ by the relation (29). Then, using the definition (26) of the dimensionless parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$, (67) becomes

$$1 + \frac{\tilde{\dot{\varepsilon}}_0^2}{12N^2} = \frac{8\sqrt{2}}{3}N\tilde{\ell}. \tag{68}$$

Moreover if one rescales the number of fragments by setting $\tilde{N} = N\tilde{\ell}$, it turns out that the rescaled number of fragments depends only on the dimensionless parameter $\eta = \tilde{\dot{\varepsilon}}_0\tilde{\ell}$:

$$1 + \frac{\eta^2}{24\tilde{N}^2} = \frac{8\sqrt{2}}{3}\tilde{N}. \tag{69}$$

4.4.4 Conclusion

One can compare the number of fragments predicted by (69) with those obtained by the numerical simulations or the stability analysis based on the gradient damage models. For that, let us distinguish two cases according to whether η is small or not.

1. *When η is small with respect to* 1 (Let us recall that this is the most frequent situation in practice, η being not small only when the strain rate is very large.) Ones deduces from (69) that $\tilde{N} \approx 3/8\sqrt{2} \approx 0.265$ and hence the number of fragments is practically independent of the strain rate. This property which comes from the simplified analysis of Glenn and Chudnovsky (1986) is in perfect agreement with the results that we have obtained as well

by the numerical simulations as by the stability analysis based on the gradient damage model. Of course the constant value 0.265 of $\tilde{N}$ coming from (69) is different from the one given by the damage model (one finds $\tilde{N} = 1/2\pi \approx 0.159$), but this discrepancy is due to the rather rough approximations made in Glenn and Chudnovsky (1986) on the various energies involved during the fragmentation process. Let us note however that this independency of the number of fragments on the strain rate is due to the assumption (made both in the simplified approach and in the damage model) that the critical stress, the toughness and hence the characteristic length of the material do not depend on the strain rate. As it is pointed out in Zinszner et al. (2015), this is not clearly the case for the alumina and more generally for the ceramics at high strain rates. If one assumes that the critical stress grows with the strain rate whereas the toughness and the Young modulus remain constant, then ℓ is increasing and ε_c is decreasing when the strain rate is increasing. Therefore, since η will decrease and hence will remain small, the number of fragments will increase like $\sqrt{\sigma_c}$ when $\dot{\varepsilon}_0$ will increase.

2. *When η is not small.* Then, still by using (68) or (69), the dependence of the number of fragments on the strain rate becomes effective. Moreover one recovers the power law of Grady (1982), $N \sim \dot{\varepsilon}_0^{2/3}$, when η is large with respect to 1. One observes the same tendency with the gradient damage model but a quantitative comparison between the two approaches is less relevant. Indeed, as one can see in Fig. 4.3, the value α_1 of the damage at which the homogeneous solution becomes unstable grows rapidly with η and is practically equal to 1 when $\eta = 0.1$. Consequently, one can no more use the previous estimates of the various energy at the moment t_1 when the fragmentation occurs: the remaining elastic energy in the ring just before t_1 is less than $\mathcal{U}^-$ (and even is close to 0 when $\eta = 0.1$), a non negligible energy has been dissipated to damage the material until α_1, the energy that is needed to create a fragment is less than $G_c S$ because the damage has only to grow from α_1 to 1 (and even that surface energy is close to 0 when $\eta = 0.1$). One could adapt the energy estimates, but that requires to know the right instant t_1. In the gradient damage approach, t_1 corresponds to the loss of stability of the uniform response. In Glenn and Chudnovsky (1986) it was assumed that $t_1 = \varepsilon_c/\dot{\varepsilon}_0$ and a criterion is missing to change this value.

5 Number of cracks in a multidimensional ring

The last part of this work consists of comparing the results obtained for the 1D periodic ring to those obtained for a 2D or 3D ring. More precisely, we want to know if the number of cracks we obtained in 1D or 3D are the same. The multidimensional problem is clearly more complex. In the 1D ring, we had two main dimensionless parameters $\tilde{\ell}$ and $\tilde{\dot{\varepsilon}}_0$. For a complete 3D ring, we also have to study the influence of the thickness, the amplitude and the form of the applied loading, and there are more possible options for the shape of the initial perturbation. Now, denoting by R_i and R_e the inner and outer radius of the ring, the main parameters are the following ones:

$$L = 2\pi R_i, \quad \tilde{\ell} = \frac{\ell}{2\pi R_i}, \quad \tilde{\dot{\varepsilon}}_0 = \frac{2\pi R_i \dot{\varepsilon}_0}{c_0 \varepsilon_c},$$

where ℓ, c_0 and ε_c are still the material constants introduced in Sect. 2. (Now, the Poisson ratio ν will also play a role.) After running a set of simulations, we obtained answers for some of these questions: the thickness has little influence on the number of cracks, as long as it is small compared to the internal radius (but not too small because very thin rings pose numerical difficulties to account for damage localization); the 2D and the 3D ring behave in the same way, so most of the tests were done in a 2D scenario. For the loading, we have considered an imposed radial displacement at the internal surface. The form of the given displacement profile at the inner surface (that is, uniform or almost uniform with sufficiently small perturbations) has little influence on the number of cracks; it does, however, play an important role in facilitating damage localization and the instant when these damage profiles appears. We tried different values for the ratio R_e/R_i of radii between 1.05 and 1.2, and obtained similar results. The ring is in its natural configuration without velocity at the initial time $\mathbf{u}_0 = \dot{\mathbf{u}}_0 = 0$. The external boundary is free of forces ($\boldsymbol{\sigma}\mathbf{e}_r = \mathbf{0}$ at $r = R_e$) and a displacement proportional to the time is imposed at the inner boundary ($\mathbf{u} = \dot{\varepsilon}_0 t \mathbf{e}_r$) for $t > 0$. We have only considered dimensionless small strain rates. For larger values of the strain rate (close or larger than one), the localization of damage is not guaranteed or is very close to

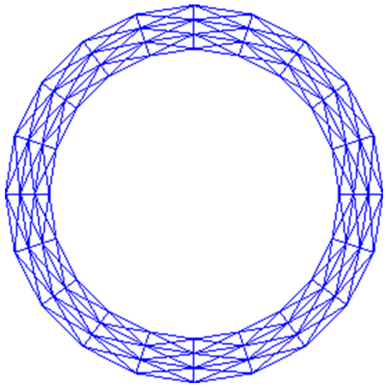

Fig. 7 Illustration of a generic mesh used for the simulations in 2D

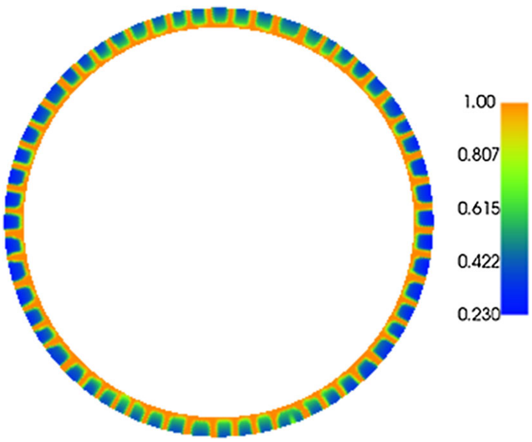

Fig. 8 Damage profile for $\dot{\varepsilon} = 3.1$ and $\tilde{\ell} = 0.0025$. We obtained 58 cracks (64 expected)

$\alpha = 1$, making it difficult to distinguish the fragments. When running our simulations, we used a structured mesh, the nodes being aligned with the direction of the motion and allowing us to better represent the dynamics of the system. Different mesh sizes were used, giving us close (with a difference of less than 20%) number of fragments. A typical mesh is illustrated in Fig. 7.

For the 1D problem, we showed that the number of cracks N^* was close to $L/(2\pi\ell)$ and that the strain rate has only a small influence on the number of fragments that appear. In the 2D setting, one still compares the computed number of fragments with N^* given by (58) (with $S = 10^{13}$), recalling that now $\tilde{\ell} = \ell/2\pi R_i$.
In each case, the spatial repartition of the fragments is practically periodic as one can see on Fig. 8. Table 1 shows the results we obtained for a fixed strain rate $\tilde{\dot{\varepsilon}}_0 = 3.1$ and different values of internal length $\tilde{\ell}$. The number of cracks estimated by N^* is rather good, even if N^* overestimates the number of cracks for lower values of ℓ, and, underestimates the number of cracks for higher values of ℓ. We used a mesh of size 10^{-3}.

Table 1 Influence of the internal length in the number of cracks for $\tilde{\dot{\varepsilon}}_0 = 3.1$

$\tilde{\ell}$	N^*	Computed number of fragments
0.01	16	23
0.005	32	38
0.0025	64	58

Table 2 Influence of the strain rate in the number of cracks for $\tilde{\ell} = 0.01$

$\dot{\varepsilon}$	N^*	Computed number of fragments
0.16	16	14
0.31	16	14
0.79	17	17
1.6	18	20
3.1	20	23

The strain rate plays a more important role in the computed number of cracks than the one predicted by N^*. For $\tilde{\ell} = 0.01$, we can see in Table 2 a significant change in the number of fragments when $\tilde{\dot{\varepsilon}}_0$ varies from 0.31 to 3.1. For values lower than $\tilde{\dot{\varepsilon}}_0 = 0.31$, we did not see a significant change in the number of cracks. For values higher than $\tilde{\dot{\varepsilon}}_0 = 3.1$, the localization of cracks was less evident and the results were not considered.

This means that the 2D and 3D cases are more complex and that the study of the one-dimensional bar is not enough to accurately predict the number of cracks. We have, however, the same tendency: an increase of the number of fragments for smaller internal lengths and higher strain rates; a number of fragments of the same order as the predicted value by the (1D) Linear Analysis of Stability.

6 Conclusion

The present paper has shown that gradient damage models are able to account for the nucleation and the multiple fragmentation of brittle materials in a dynamical setting, opening thus a large domain of applications. Their use for treating the delicate problem of the fragmentation of a brittle ring under a uniform expansion at relative high strain rate allows us to better understand what are the reasons which lead to the nucleation of cracks from a homogeneous response. Although it seems that the dynamical damage evolution problem admits a unique solution, which excludes the possibility of a bifurcation contrarily to its quasi-static counterpart, a linear analysis of stability has shown that the nucleation of cracks is governed by the growth of imperfections. Moreover, this stability analysis has allowed us to obtain in a closed form the dependence of the number of fragments on the parameters of the problem. In particular one has obtained that the characteristic length entering in the used gradient damage model plays the most essential role to work out the number of fragments, the role of the strain rate being weaker, as long as the strain rate is small enough. Specifically, one has shown that the number of fragments is essentially function of the ratio ℓ/L between the characteristic length of the material and the length of the ring as long as the strain rate $\dot{\varepsilon}_0$ is much smaller than the material parameter $c_0\varepsilon_c/\ell$ involving the speed of the elastic waves, the critical strain and the characteristic length of the material. But, when the strain rate becomes large with respect to $c_0\varepsilon_c/\ell$, the fact that the homogeneous response becomes unstable only when the damage is close to 1 is clearly a weakness of the present gradient damage model. One should improve it (may be by adding rate effects) in order that the loss of stability occurs when the ring is weakly damaged even when the strain rate is large.

The unique purpose of the present work was to better understand the behavior of gradient damage models in dynamics without any real confrontation with experimental results. Of course, such a comparison is absolutely necessary to be sure that the models in their present form are sufficient to predict with a good accuracy the number of fragments for a large range of the strain rate. Future works will be devoted to this fundamental task.

A Asymptotic analysis of the differential equation governing the amplitude of the perturbations

Let us consider the second order linear differential equation

$$\eta^2\ddot{y}(t) = f(t)y(t) \tag{70}$$

where η is a small parameter and the dot indicates the derivative with respect to t. In (70) f is a smooth function defined for $t > 0$ and such that

$$f(t)\begin{cases} < 0 & \text{if} \quad 0 < t < t_c \\ = 0 & \text{if} \quad t = t_c \\ > 0 & \text{if} \quad t > t_c \end{cases}, \qquad \dot{f}(t_c) > 0.$$

In the application we will take

$$f(t) = \frac{3t^2 - 1}{t^4(t^2+1)} \tag{71}$$

and in such a case $t_c = 1/\sqrt{3}$, $\dot{f}(t_c) = \sqrt{3}/6$. Denoting the general solution by y_η, one wants to construct an asymptotic approximation of y_η. For that one must discriminate three different cases according to t is smaller than, close to or greater than the zero of f, t_c.

1. *When* $t > t_c$. Then, setting $y(t) = \exp(z(t)/\eta)$, the differential equation (70) becomes

   $$\eta\ddot{z}(t) + \dot{z}(t)^2 - f(t) = 0. \tag{72}$$

   The general solution z_η of (72) can be expanded in powers of η:

   $$z_\eta(t) = z_0(t) + \eta z_1(t) + \eta^2 z_2(t) + \cdots.$$

   Inserting that expansion into (72) gives for the first two terms:

   $$\dot{z}_0(t)^2 = f(t), \quad \ddot{z}_0(t) + 2\dot{z}_0(t)\dot{z}_1(t) = 0.$$

   One easily deduces that $z_0(t) = \pm F(t)$ where F is defined on $(0, +\infty)$ by

   $$F(t) = \left| \int_{t_c}^{t} \sqrt{|f(s)|}ds \right| \tag{73}$$

   and $z_1(t) = -\frac{1}{4}\ln f(t)+\text{cst}$. Therefore the general solution y_η can be approximated (up to the "second order") by

   $$y_\eta(t) = \frac{a_1 \exp\left(-\dfrac{F(t)}{\eta}\right) + b_1 \exp\left(\dfrac{F(t)}{\eta}\right)}{f(t)^{1/4}} \tag{74}$$

   where a_1 and b_1 are two arbitrary constants. (Note that we could refine the approximation by calculating the next terms of the expansions. However, the present approximation up to second order will be sufficient for our purpose.)

2. *When* $0 < t < t_c$. Then, setting $y(t) = \exp(\mathrm{i}z(t)/\eta)$ where i is the complex number equal to $\sqrt{-1}$, one obtains by the same procedure that the general solution y_η can be approximated (up to the "second" order") by

$$y_\eta(t) = \frac{a_0 \sin\left(\frac{F(t)}{\eta} + \frac{\pi}{4}\right) + b_0 \cos\left(\frac{F(t)}{\eta} + \frac{\pi}{4}\right)}{|f(t)|^{1/4}} \tag{75}$$

where a_0 and b_0 are two arbitrary constants and $F(t)$ is still given by (73). Note that the exponential functions of the previous case are here replaced by sinusoidal functions because of the change of sign of f at $t = t_c$.

3. *In the neighborhood of* t_c. When t tends to t_c from below or from above, the approximations (75) and (74) diverge because $f(t_c) = 0$. One must construct another approximation when t is close to t_c and match it to the two previous ones. Note that, when t is close to t_c, $f(t) \approx \dot{f}(t_c)(t - t_c)$ and $F(t) \sim \frac{2}{3}\sqrt{\dot{f}(t_c)}|t - t_c|^{3/2}$. So, in order that $F(t)$ be of the order of η, one makes the following rescaling of t:

$$t = t_c + \eta^{2/3} T,$$

T being the rescaled time. Then, using the approximation $f(t) \approx \dot{f}(t_c)(t - t_c)$, the differential equation (70) becomes

$$\frac{d^2 y}{dT^2} = \dot{f}(t_c) T y.$$

Its general solution involves the two Airy functions $\mathsf{A_i}$ and $\mathsf{B_i}$ defined by the following indefinite integrals:

$$\mathsf{A_i}(t) = \frac{1}{\pi}\int_0^{+\infty} \cos\left(\frac{s^3}{3} + ts\right) ds,$$

$$\mathsf{B_i}(t) = \frac{1}{\pi}\int_0^{+\infty} \left(\exp\left(-\frac{s^3}{3} + ts\right) + \sin\left(\frac{s^3}{3} + ts\right)\right) ds.$$

Specifically, the general solution y_η of (70) can be approximated by

$$y_\eta(t_c + \eta^{2/3} T) = a_* \mathsf{A_i}\left(\dot{f}(t_c)^{1/3} T\right) + b_* \mathsf{B_i}\left(\dot{f}(t_c)^{1/3} T\right), \tag{76}$$

where a_* and b_* are two arbitrary constants.

In order that the three asymptotic approximations correspond to the same function y_η on the half real line, one must adjust the six constants. For that, one must write the matching conditions when $|t - t_c|$ is small but $|T|$ is large. On one hand, one uses the fact that

$$F(t_c + \eta^{2/3} T) \approx \frac{2}{3}\eta\sqrt{\dot{f}(t_c)}\, T^{3/2},$$

$$f(t_c + \eta^{2/3} T) \approx \eta^{2/3} \dot{f}(t_c)\, T,$$

and, in the other hand, one uses the following behaviors of the Airy functions near $\pm\infty$:

$$\mathsf{A_i}(t) \approx \begin{cases} \dfrac{\exp\left(-\frac{2}{3} t^{3/2}\right)}{2\sqrt{\pi} t^{1/4}} & \text{when} \quad t \to +\infty \\ \dfrac{\sin\left(\frac{2}{3}|t|^{3/2} + \frac{\pi}{4}\right)}{\sqrt{\pi}|t|^{1/4}} & \text{when} \quad t \to -\infty \end{cases},$$

$$\mathsf{B_i}(t) \approx \begin{cases} \dfrac{\exp\left(\frac{2}{3} t^{3/2}\right)}{\sqrt{\pi} t^{1/4}} & \text{when} \quad t \to +\infty \\ \dfrac{\cos\left(\frac{2}{3}|t|^{3/2} + \frac{\pi}{4}\right)}{\sqrt{\pi}|t|^{1/4}} & \text{when} \quad t \to -\infty \end{cases}.$$

Comparing (74) and (75) with (76), one gets

$$a_0 = \left(\eta \dot{f}(t_c)\right)^{1/6} \frac{a_*}{\sqrt{\pi}} = 2a_1$$

$$b_0 = \left(\eta \dot{f}(t_c)\right)^{1/6} \frac{b_*}{\sqrt{\pi}} = b_1.$$

Therefore, the general solution of (70) can be approximated by

$$y_\eta(t) = \begin{cases} \left(2a_1 \sin\left(\frac{F(t)}{\eta} + \frac{\pi}{4}\right) + b_1 \cos\left(\frac{F(t)}{\eta} + \frac{\pi}{4}\right)\right) |f(t)|^{-1/4} & \text{when} \quad 0 < t < t_c - O(\eta^{2/3}) \\ \dfrac{2a_1\sqrt{\pi}}{\eta^{1/6}\dot{f}(t_c)^{1/6}} \mathsf{A_i}\left(\dot{f}(t_c)^{1/3}\dfrac{t - t_c}{\eta^{2/3}}\right) + \dfrac{b_1\sqrt{\pi}}{\eta^{1/6}\dot{f}(t_c)^{1/6}} \mathsf{B_i}\left(\dot{f}(t_c)^{1/3}\dfrac{t - t_c}{\eta^{2/3}}\right) & \text{when} \quad |t - t_c| \le O(\eta^{2/3}) \\ \left(a_1 \exp\left(-\frac{F(t)}{\eta}\right) + b_1 \exp\left(\frac{F(t)}{\eta}\right)\right) f(t)^{-1/4} & \text{when} \quad t > t_c + O(\eta^{2/3}) \end{cases}, \tag{77}$$

the two constants a_1 and b_1 being fixed by initial conditions.

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